

AFRICA GRID ANALYTICS LANDSCAPE

A deep competitive analysis of global and African grid analytics players — with identified market gaps and actionable hardware & software opportunities for the next wave of energy infrastructure investment.

600M

Africans without electricity

49.5%

Avg ATC&C losses, Nigeria

\$10B

Potential annual savings by 2030

\$65B

Mini-grid investment needed

// 01 - COMPETITIVE LANDSCAPE

COMPETITOR PROFILES

GLOBAL ENTERPRISE TIER

SCHNEIDER ELECTRIC

Operates the One Digital Grid Platform (launched 2023) with EcoStruxure ADMS, DERMS, and Net Zero Dashboard. Middle East & Africa segment grew 19% in 2024. Targeting grid resilience, DER integration, and sustainability metrics.

ADMS

DERMS

IOT

Digital Twin

Africa Active

STRENGTHS

→ Proven global deployment

→ 40% outage reduction claimed

→ DER interconnection tools

→ Enterprise-grade security

GAPS IN AFRICA

→ High licensing costs

→ Not tailored to LV networks

→ Requires stable connectivity

→ Enterprise sales cycle too slow

GLOBAL ENTERPRISE TIER

SIEMENS GRID SOFTWARE

Enterprise-grade digital twin and predictive analytics for massive utility networks. Partners with renewable energy companies (wind, solar) to integrate grid management at scale. Global standard for transmission-level analytics.

Digital Twin

Predictive AI

Transmission

Limited Africa

STRENGTHS

→ Transmission-scale analytics

→ Deep simulation capability

→ Established global partnerships

GAPS IN AFRICA

→ Minimal distribution-level tools

→ Near-zero rural/mini-grid play

→ Data requirements impractical

→ No local language/currency support

AFRICA-NATIVE TIER

BEACON POWER SERVICES

Accra-based energy tech. CAIMS (GIS/asset mapping) and ADORA (real-time grid monitoring). Raised Series A in Nov 2024, led by Partech. Active in Ghana (ECG), Nigeria, Tanzania, Zambia, Togo. Won 2025 PMI State Organization Project of the Year.

GIS Mapping

Real-time SCADA

Revenue Protection

Ghana/Nigeria/Togo

STRENGTHS

→ Built for Africa's realities

→ Boots-on-ground approach

→ Proven ECG digitization

→ In-house development

GAPS

→ No hardware manufacturing

→ No mini-grid/off-grid product

→ Limited renewable forecasting

→ No consumer-facing tools

AFRICA-ADJACENT TIER

AWESENSE

Canadian grid intelligence company with Africa market positioning. Focuses on LV network analytics, loss detection, and grid viability for utilities. Positioned alongside BPS as a market leader for localized loss reduction.

LV Analytics

Loss Detection

Non-native

STRENGTHS

→ Strong LV network analytics

→ Non-technical loss detection

→ Modular deployment

GAPS

→ Foreign pricing model

→ Limited local ecosystem

→ Needs existing metering infra

PLATFORM / ADJACENT TIER

HITACHI ENERGY (LUMADA)

Lumada IoT platform with grid analytics, transformer monitoring, and AI-driven efficiency tools. Showcased Africa-specific solutions at ADIPEC 2024. Strong in ME&A on hardware (transformers) but analytics are platform-heavy.

IoT Platform

Transformer Monitoring

Edge AI (2025)

STRENGTHS

→ Hardware + software play

→ Edge AI development (2025)

→ Established Africa hardware

GAPS

→ Platform lock-in risk

→ Not distribution-utility focused

→ Expensive integration overhead

FINANCE / INFRASTRUCTURE TIER

AFRICA ENERGY MGMT PLATFORM (AEMP)

Develops, finances, and operates utility-scale renewable and hybrid energy assets. Not a software analytics company — focuses on capital allocation and project finance for clean energy. Different value chain position.

Project Finance

Renewable Assets

Hybrid Systems

STRENGTHS

→ Capital + operations combined

→ Utility-scale renewable focus

GAPS

→ Not a software/analytics player

→ No real-time grid visibility

→ No distribution utility tools

CAPABILITY MATRIX

PLAYER	LV NETWORK ANALYTICS	MINI-GRID SUPPORT	AFRICA-NATIVE DATA	OFFLINE/EDGE MODE	CONSUMER TOOLS	HARDWARE	REVENUE MODEL
Schneider Electric	● Partial	● No	● No	● Limited	● DERMS only	● Yes	SaaS + Hardware
Siemens Grid Software	● No	● No	● No	● No	● No	● Yes	Enterprise License
Beacon Power Services	● Yes	● No	● Yes	● Partial	● No	● No	SaaS + Data Services
Awesense	● Yes	● No	● No	● Partial	● No	● No	SaaS
Hitachi Lumada	● Partial	● No	● No	● Edge AI 2025	● No	● Yes	Platform + Hardware

CRITICAL GAPS IDENTIFIED

01

No Offline-First Analytics

All major platforms assume reliable internet. In 2025, most African utilities still manually operate analog systems. None of the profiled players offer a grid analytics stack that works meaningfully offline or on low-bandwidth GPRS connections.

02

Mini-Grid Intelligence Vacuum

Africa needs 20,000–40,000 new mini-grids per year to hit SDG7. Yet not one of the profiled competitors has a purpose-built analytics product for mini-grid operators. Mini-grids are managed with spreadsheets and guesswork.

03

No Affordable Smart Meter Hardware

Smart metering hardware in Africa still requires import financing and international supply chains. No African company manufactures locally adapted, solar-powered, tamper-resistant meters at volume. BPS itself relies on imported devices.

04

Revenue Leakage Goes Unmeasured

Nigeria's ATC&C losses run 40–50%. Collection efficiency at 76.3% means utilities bill but don't collect. No platform currently combines GIS mapping, smart metering, and AI payment prediction to proactively identify and close collection gaps.

05

No Consumer-Facing Energy Apps

Electricity customers in Africa have zero visibility into their own usage, bill predictions, or outage schedules. In markets with prepaid meters, consumers need real-time balance, consumption alerts, and credit top-up integrations — none exist at scale.

06

Renewable Intermittency Management

Africa's renewable share is growing rapidly — 76% of electricity from renewables projected by 2040. But no Africa-native tool forecasts solar/wind intermittency at the distribution level or automates battery dispatch for hybrid mini-grids in real time.

07

No Grid Asset Lifecycle Platform

Aging transformer banks, overhead lines, and switchgear fail unpredictably across the continent. Global players offer predictive maintenance only for high-value transmission assets. Distribution-level asset health monitoring for African conditions doesn't exist locally.

08

Fragmented Data, No Interoperability

Utilities in the same country use incompatible SCADA, billing, and GIS systems. There is no Africa-specific grid data standard or interoperability layer. Utilities cannot benchmark against peers or aggregate national grid performance intelligence.

HARDWARE & SOFTWARE IDEAS



HARDWARE IDEA — #1

GRIDPULSE NODE

Low-cost, solar-powered distribution monitoring unit built for African LV networks — works offline, communicates via SMS/GPRS

WHAT IT DOES

- Clamps onto transformer feeders, LV busbar, or service drops
- Measures voltage, current, power factor, harmonics every 15 min
- On-device edge AI flags fault signatures and theft patterns
- Solar-charged battery, 30-day autonomy
- SMS-based reporting when no data connection available
- Tamper-evident casing with GPS tag

REVENUE MODEL

- Hardware sale to utilities: \$80–\$120/unit at volume
- \$5/unit/month cloud data subscription
- Integration fee with BPS ADORA or ECG SCADA: \$50k/utility
- Carbon credit revenue via measurable loss reduction
- DFI/grant funding as first-mover cost recovery

TAM: ~500k transformer feeders across sub-Saharan Africa

IMPACT

- Reduces outage detection time from hours to minutes
- Identifies 20–35% of non-technical losses at the feeder level
- Enables revenue recovery of \$200k–\$1M/utility/year
- Works without smartphones or reliable internet
- Designed and assembled locally — creates manufacturing jobs

Addresses Gap #1, #3, #4



HARDWARE IDEA — #2

AFRIMETER PRO

Africa-first prepaid smart meter with built-in mobile money integration and tamper-resistant NB-IoT communication

WHAT IT DOES

- Prepaid meter with NB-IoT + fallback SMS
- Direct integration with MTN MoMo, M-PESA, Airtel Money for top-up
- Customer balance alerts via USSD (no smartphone needed)
- Tamper sensor with auto-alert and remote disconnect
- Local display shows balance, usage, and tariff
- Anti-bypass housing — key gap in current meter designs

REVENUE MODEL

- Hardware: \$35–\$55/unit to utilities (vs \$80–\$120 for imports)
- Mobile money payment processing fee: 0.3–0.5% per transaction
- SaaS meter data management platform: \$2/meter/month
- White-label sales to private landlords and industrial parks
- IFC/USAID subsidy programs eligible

Market: 60M+ un-metered connections in SSA

IMPACT

- Directly closes billing & collection efficiency gap
- Mobile money integration boosts payment rates 25–40%
- Reduces meter reading cost by 100% (automated)
- Localizes a \$1B+ import market into African manufacturing
- Empowers customers with real-time usage data via USSD

Addresses Gap #3, #4, #5



SOFTWARE IDEA — #1

VOLTIQ — AI LOSS INTELLIGENCE PLATFORM

Africa-native SaaS that turns raw meter, SCADA, and GIS data into revenue recovery actions — with built-in offline sync

WHAT IT DOES

- Ingests data from any meter brand, SCADA, or manual reading
- AI engine classifies each loss event: technical vs commercial vs theft
- Generates field crew dispatch orders ranked by revenue impact
- Offline-first PWA: field agents work without internet, syncs on reconnect
- Benchmarks feeder-level performance against national peers
- Multilingual: English, French, Swahili, Hausa, Twi

REVENUE MODEL

- SaaS: \$0.50/customer/month (typical utility + 200k customers = \$100k ARR)
- Performance-based tier: 10% of recovered revenue in Year 1
- Data insights API: sell anonymized grid health data to DFIs & regulators
- Training & change management services: \$20–\$50k/engagement
- Carbon credit accounting module (premium): \$5k/year/utility

Potential ARR: \$5–20M across 20 utilities

IMPACT

- 10–20% ATC&C loss reduction in first 12 months
- \$1–5M revenue recovery per mid-size utility per year
- Reduces dependence on expensive international consultants
- Works on 2G/3G and fully offline for rural substations
- Creates data-driven accountability between utilities and regulators

Addresses Gap #1, #4, #8



SOFTWARE IDEA — #2

MINIOPS — MINI-GRID OPERATING SYSTEM

The first purpose-built management platform for Africa's 200,000+ planned mini-grids: generation forecasting, battery dispatch, and community billing

WHAT IT DOES

- Real-time solar PV generation forecasting (15-min intervals)
- Battery energy storage automated dispatch AI
- Community energy budget management — allocate fairly across households
- Mobile money billing with USSD interface for customers

REVENUE MODEL

- \$200/mini-grid/month SaaS subscription
- Transaction fee: 0.1% on all mobile money payments processed
- Carbon credit MRV (measurement, reporting, verification): \$5k/site/year
- Sell anonymized demand data to solar manufacturers and DFIs

IMPACT

- Cuts mini-grid O&M cost by 30% via predictive scheduling
- Improves battery asset life by 15–20% via optimal dispatch
- Unlocks carbon credit revenues for mini-grid operators
- Provides investors with real-time project performance data

- Technician app for fault logging and maintenance scheduling
- Investor dashboard: remote financial performance monitoring

- White-label for World Bank / AfDB mini-grid programs

TAM: 110,000 planned mini-grids by 2030 = \$264M ARR

- Bridges the \$6B mini-grid investment gap with data credibility

Addresses Gap #2, #6

 SOFTWARE IDEA - S3

POWERPAL – CONSUMER ENERGY APP

Africa's first consumer-facing electricity intelligence app: usage tracking, outage alerts, bill prediction, and peer-to-peer energy sharing

WHAT IT DOES

- Real-time electricity balance and usage tracking via smart meter API
- Hyperlocal outage forecasting based on utility data + crowd reports
- Bill prediction: "you'll need to top up \$8 by Thursday"
- Tariff comparison across zones and time-of-use options
- Appliance energy audit: AI estimates usage by appliance type
- PaP rooftop solar credit marketplace (prosumer model)

REVENUE MODEL

- Freemium app: free basic, \$1.5/month premium
- Utility white-label: \$50k/year per utility for branded version
- Embedded mobile money top-up: 0.5% transaction fee
- Appliance brand advertising: "your fridge uses 40kWh/month; upgrade here"
- Energy efficiency lending partner referrals: \$10-\$30/conversion

Target: 5M users in 3 years = \$7.5-15M ARR

IMPACT

- First product to give 100M African customers grid visibility
- Demand response enablement at consumer scale
- Reduces peak-load-related outages via behavioral nudges
- Creates data layer for utilities to understand real demand patterns
- Enables prosumer participation – democratizes the energy transition

Addresses Gap #5

 HARDWARE + SOFTWARE IDEA - H51

GRIDBRIDGE – INTEROPERABILITY HUB

A plug-in hardware gateway + open API middleware that connects legacy SCADA, imported meters, and GIS databases into a single standardized African grid data layer

WHAT IT DOES

- Hardware gateway: connects to any RS485, Modbus, or DLMS meter protocol
- Translates incompatible data formats into IEC 61968/61970 standard
- Open API layer: BPS, Awesense, or custom dashboards can connect
- National-level grid intelligence aggregation without replacing existing systems
- Regulators can access benchmarks without utility vendor access

REVENUE MODEL

- Hardware gateway: \$500-\$1,500 per substation
- API subscription: \$2,000/month per utility
- National regulator data dashboard: \$100k/year contract
- Data marketplace for energy researchers and DFIs: \$50k/year
- System integration services: \$200-\$500k per large utility rollout

B2B SaaS + Regulatory contracts

IMPACT

- Unlocks data trapped in 20+ incompatible legacy systems
- Enables continent-wide grid benchmarking for the first time
- Reduces vendor lock-in for African utilities
- Creates an open standard – network effects grow the ecosystem
- Powers evidence-based energy regulation across Africa

Addresses Gap #7, #8

// STRATEGIC SYNTHESIS

The dominant pattern across all profiled competitors is a **top-down, enterprise-first model** designed for well-capitalized Western utilities. Africa's energy crisis – 600M unconnected, 40–50% losses, analog grid operations, and 600M people paying for backup generators – is fundamentally a **distribution and data problem**, not a generation problem. The highest-impact opportunity is a vertically integrated Africa-native company that combines **affordable locally-manufactured hardware** (GridPulse Node, Afrimeter Pro) with **offline-capable, AI-powered analytics software** (Voltiq, MiniOps) and a **consumer-facing energy intelligence layer** (PowerPal) – creating a full-stack grid intelligence ecosystem that none of the current competitors has built for this market.